How-To Library

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Configuring AWS IAM Authentication for PowerExchange for Amazon Redshift

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Abstract

You can use AWS Identity and Access Management (IAM) to control individual and group access to Amazon Redshift resources. You can configure AWS IAM to run mappings on the Data Integration Service that is installed on the EC2 system. This article describes the guidelines to configure IAM Authentication for PowerExchange for Amazon Redshift.

Supported Versions

- PowerExchange® for Amazon Redshift 10.2

Table of Contents

Overview. 2
Create a Minimal Amazon S3 Bucket Policy. 2
Create the Amazon EC2 Role. 3
Create the Amazon Redshift Role. 3
Associate the Amazon Redshift Role with the Redshift Cluster. 4
Create the Amazon EMR Role. 4
Creating an Amazon Redshift Connection. 5

Overview

To control the access of Amazon Redshift resources, you can define permissions to the users by configuring AWS Identity and Access Management (IAM). The AWS IAM service provides enhanced security.

Perform the following steps to configure IAM authentication:

1. Create a minimal Amazon S3 bucket policy.
2. Create the Amazon EC2 Role.
3. Create the Amazon Redshift Role.
4. Create the Amazon EMR Role.

   Note: To use AWS IAM authentication to run a mapping in the EMR cluster, you must create the Amazon EMR Role. Then, assign the Amazon ERM Role and Amazon Redshift Role to the EMR cluster.
5. Associate the Amazon Redshift Role with the Redshift cluster.
6. Create an Amazon Redshift connection.

Create a Minimal Amazon S3 Bucket Policy

The minimal Amazon S3 bucket policy restricts user operations and user access to a particular Amazon S3 bucket by assigning an AWS IAM policy to the users. You can configure the AWS IAM policy through the AWS console.

You can use the following minimum required actions when you use PowerExchange Amazon Redshift to successfully read data from and write data to Amazon Redshift resources:

- PutObject
- GetObject
- DeleteObject
- ListBucket

- GetBucketPolicy

The following snippet shows a sample Amazon S3 bucket policy for PowerExchange Amazon Redshift:

```
{
"Version": "2012-10-17", "Statement": [{ "Effect": "Allow", "Action": [ "s3:PutObject",
"s3:GetObject", "s3:DeleteObject", "s3:ListBucket", "s3:GetBucketPolicy" ], "Resource":
[ "arn:aws:s3:::<specify_bucket_name>/*", "arn:aws:s3:::<specify_bucket_name>" ] }
]
}
```

Create the Amazon EC2 Role

You must create an Amazon EC2 Role to provide users access to the Redshift resources. You can use the Amazon EC2 Role when you create an EC2 system in the Amazon Redshift cluster.

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel.

   The **AWS Service** dashboard page appears.
3. Click **IAM**.

   The **Welcome to Identity and Access Management** page appears.
4. Click **Policies** from the left panel.

   The **Policy** page appears.
5. Click **Create Policy** or select the required existing Amazon S3 Policy.

   You can edit or review the policy.
6. Select **Role** from the left panel and click **Create role**.

   The **Create role** page appears.
7. Select **EC2** under the **Choose the service that will use this role** section.
8. Select the required **Amazon EC2** role type under the **Select your use case** section.
9. Click **Next: Permission**.
10. Select the required **Amazon S3 Policy** in the **Attach Permission Policies** page.
11. Click **Next: Review**.
12. Specify the name of the role in the **Create role** review page.
13. Click **Create Role**.
14. Review the Role ARN, Instance Profile ARNs, and Policy values in the **Summary** page.

After you create the Amazon EC2 Role, create an EC2 instance. Assign the Amazon EC2 Role to the EC2 instance. For more information about creating an EC2 instance and assigning an Amazon EC2 Role to the Amazon EC2 instance, see the Amazon Redshift documentation.

Create the Amazon Redshift Role

You must create the Amazon Redshift Role Resource Name (ARN) for secure access to Amazon Redshift resources.

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel.

   The **AWS Service** dashboard page appears.
3. Click **IAM**.

   The **Welcome to Identity and Access Management** page appears.

4. Click **Policies** from the left panel.

   The **Policy** page appears.
5. Click **Create Policy** or select the required existing Amazon S3 Policy.

   You can edit or review the policy.
6. Select **Role** from the left panel and click **Create role**.

   The **Create role** page appears.
7. Select **Redshift** under the **Choose the service that will use this role** section.
8. Select the required Amazon Redshift cluster under the **Select your use case** section.
9. Click **Next: Permission**.
10. Select the required **Amazon S3 Policy** in the **Attach Permission Policies** page.
11. Click **Next: Review**.
12. Specify the name of the role in the **Create role** review page.
13. Click **Create Role**.
14. Review the Role ARN, Instance Profile ARNs, and Policy values in the **Summary** page.

The following example shows a sample Amazon Redshift Role Resource Name (ARN):

```
arn:aws:iam::123123456789:role/redshift_write
```

You must assign the Amazon Redshift Role that you created to the Amazon Redshift cluster to successfully perform the read and write operations.

Associate the Amazon Redshift Role with the Redshift Cluster

You must associate the Amazon Redshift Role Resource Name (ARN) with an Amazon Redshift cluster to read data from Amazon Redshift and write data to the Amazon S3 bucket.

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel.

   The **AWS Service** dashboard page appears.
3. Click **Amazon Redshift**.

   The **Redshift** dashboard page appears.
4. Click **Clusters** under **Resources**.
5. Select the required Amazon Redshift cluster.
6. Click **Manage IAM roles**.

   The **Manage IAM roles** dialog box displays.
7. Select the required Amazon Redshift Role. For example, `arn:aws:iam::123123456789:role/redshift_write`
8. Click **Apply changes**.

After you associate the Amazon Redshift Role Resource Name (ARN) with the Amazon Redshift cluster, install the Data Integration Service on the EC2 instance.

Create the Amazon EMR Role

You can use AWS IAM authentication when you run a mapping in the EMR cluster.

To use AWS IAM authentication in the EMR cluster, you must create the Amazon EMR Role. You must assign both the Amazon EMR Role and Amazon Redshift Role to the EMR cluster for secure access to Amazon Redshift resources.

Note: Before you configure IAM Role with EMR cluster, you must install the Informatica Services on an EC2 instance with the IAM Roles assigned.

1. Log in to the **AWS Console**.
2. Click **Dashboard** from the left panel.

   The **AWS Service** dashboard page appears.
3. Click **IAM**.

   The **Welcome to Identity and Access Management** page appears.
4. Click **Policies** from the left panel.

   The **Policy** page appears.
5. Click **Create Policy** or select the required existing Amazon S3 Policy.

   You can edit or review the policy.
6. Select **Role** from the left panel and click **Create role**.

   The **Create role** page appears.
7. Select **EMR** under the **Choose the service that will use this role** section.
8. Select the required EMR cluster under the **Select your use case** section.
9. Click **Next: Permission**.
10. Select the required **Amazon S3 Policy** in the **Attach Permission Policies** page.
11. Click **Next: Review**.
12. Specify the name of the role in the **Create role** review page.
13. Click **Create Role**.
14. Review the Role ARN, Instance Profile ARNs, and Policy values in the **Summary** page.

Creating an Amazon Redshift Connection

When you set up an Amazon Redshift connection, you must specify the connection properties for configuring the IAM authentication to control secure access of Amazon Redshift resources.

1. In the Developer tool, click **Window** > **Preferences**.
2. Select **Informatica** > **Connections**.
3. Expand the domain in the **Available Connections**.
4. Select the connection type **Database** > **Amazon Redshift**, and click **Add**.
5. Enter a connection name and an optional description.
6. Select **Amazon Redshift** as the connection type.
7. Click **Next**.
8. Configure the following Amazon Redshift connection properties in the **Details** tab:

Property	Description
Username	User name of the Amazon Redshift account.
Password	Password for the Amazon Redshift account.

Property	Description
Schema	Optional. Amazon Redshift schema name. Do not specify the schema name if you want to use multiple schema. The Data Object wizard displays all the user-defined schemas available for the Amazon Redshift objects.
AWS Access Key ID	Not applicable when you use AWS IAM authentication.
AWS Secret Access Key	Not applicable when you use AWS IAM authentication.
Master Symmetric Key	Optional. Provide a 256-bit AES encryption key in the Base64 format when you enable client-side encryption. You can generate a key using a third-party tool.
Customer Master Key ID	Optional. Specify the customer master key ID or alias name generated by AWS Key Management Service (AWS KMS). You must generate the customer master key corresponding to the region where Amazon S3 bucket resides. You can specify either a customer generated customer master key or a default customer master key. **Note:** You can use customer master key ID when you run a mapping in the native environment or on the Spark engine.
Cluster Node Type	Node type of the Amazon Redshift cluster. You can select the following options: - ds1.xlarge - ds1.8xlarge - dc1.large - dc1.8xlarge - ds2.xlarge - ds2.8xlarge
Number of Nodes in Cluster	Number of nodes in the Amazon Redshift cluster.
JDBC URL	Amazon Redshift connection URL.

9. Click **Test Connection** to verify the connection to Amazon Redshift.
10. Click **Finish**.

Provide the Amazon Redshift Role Resource Name (ARN) in the **AWS_IAM_ROLE** option in the **UnloadOptions Property File** and **CopyOptions Property File** properties when you create a mapping.

The Data Integration Service uses the Amazon Redshift Role Resource Name (ARN) associated with the IAM Role to access the data from the Amazon Redshift target. When you run the mapping, the Data Integration Service validates the IAM policy.

Author

Salam Subhashree

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